COS30018 - Option C - Task 1 Report

Stock Prediction System Setup and Evaluation

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Course: COS30018 - Intelligent Systems
Project: FinTech101 Stock Price Prediction System

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1. Environment Setup

1.1 Virtual Environment Configuration

A unified virtual environment was created to support both v0.1 and P1 implementations, ensuring compatibility and dependency management.

Setup Process:

```
# Create and activate virtual environment
python -m venv venv
venv\Scripts\activate  # Windows
source venv/bin/activate  # Linux/Mac

# Install dependencies
pip install -r requirements.txt
```

1.2 Requirements File Development

The requirements.txt file was developed to address dependencies for both projects:

```
# Essential packages for stock prediction project
# Core machine learning
tensorflow>=2.19.0
keras>=3.11.1
scikit-learn>=1.7.1
numpy>=2.1.3
pandas>=2.3.1

# Data visualization
matplotlib>=3.10.5

# Stock data fetching (we use yfinance, keeping yahoo-fin for compatibility)
yfinance>=0.2.65
yahoo-fin>=0.8.9.1
pandas-datareader>=0.10.0

# HTTP requests and web scraping
requests>=2.32.4
requests-html>=0.10.0
beautifulsoup4>=4.13.4
lxml>=6.0.0
lxml_html_clean>=0.4.2

# Jupyter notebook support (optional)
jupyter>=1.0.0
ipykernel>=6.30.1

# Utilities
python-dateutil>=2.9.0.post0
pytz>=2025.2
loguru>=0.7.0
```

1.3 Installation and Compatibility Issues

Packages Issues Resolved:

1. **yfinance vs yahoo_fin:** Replaced unreliable yahoo_fin with stable yfinance library
2. **TensorFlow Compatibility:** Fixed deprecated parameter (huber-loss -> huber, input to Input() layer) usage in P1
3. **lxml Dependencies:** Resolved HTML parsing compatibility issues

2. Code Testing Results

2.1 v0.1 Testing Results

Code Additions:

- Modularize P1 metrics evaluating function and reuse for v0.1 for better models' performance comparison:

```
def calculate_trading_metrics(actual_prices, predicted_prices, current_prices=None,
                              lookup_step=1, future_price=None, loss_value=None,
                              loss_name="loss", filename=None):
    """
    Calculate trading accuracy and profit metrics.

    Simple trading strategy:
    - BUY when predicted > current (profit = actual - current)
    - SELL when predicted < current (profit = current - actual)
    """
    actual_prices = np.array(actual_prices).flatten()
    predicted_prices = np.array(predicted_prices).flatten()

    if current_prices is None:
        current_prices = actual_prices
    else:
        current_prices = np.array(current_prices).flatten()

    # Calculate buy and sell profits
    buy_profits = []
    sell_profits = []

    for i in range(len(actual_prices)):
        current = current_prices[i]
        pred_future = predicted_prices[i]
        true_future = actual_prices[i]

        # Buy if predicted price increase
        if pred_future > current:
            buy_profit = true_future - current
        else:
            buy_profit = 0

        # Sell if predicted price decrease
        if pred_future < current:
            sell_profit = current - true_future
```

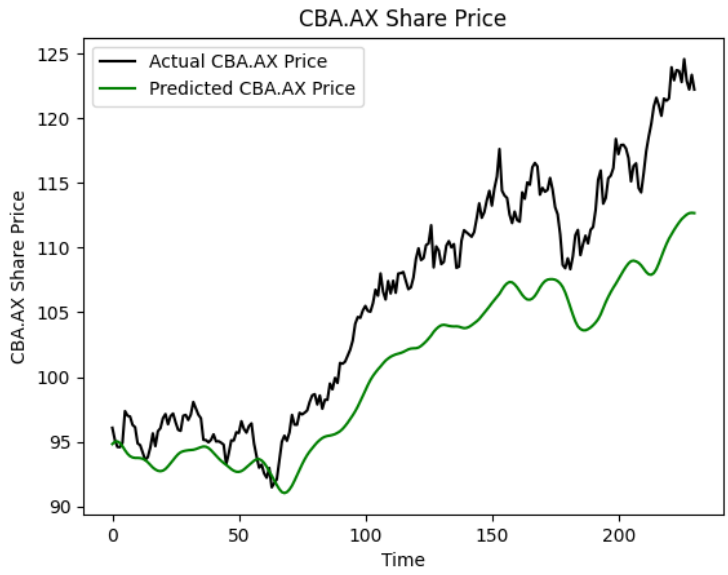
```
        else:
            sell_profit = 0

        buy_profits.append(buy_profit)
        sell_profits.append(sell_profit)

    # Calculate metrics
    total_buy_profit = sum(buy_profits)
    total_sell_profit = sum(sell_profits)
    total_profit = total_buy_profit + total_sell_profit
    profit_per_trade = total_profit / len(actual_prices)

    # Count profitable trades
    profitable_trades = sum(1 for bp, sp in zip(buy_profits, sell_profits) if bp > 0 or sp >
0)
    accuracy_score = profitable_trades / len(actual_prices)

    # Mean absolute error
    mean_absolute_error = np.mean(np.abs(actual_prices - predicted_prices))

    results = {
        'accuracy_score': accuracy_score,
        'total_buy_profit': total_buy_profit,
        'total_sell_profit': total_sell_profit,
        'total_profit': total_profit,
        'profit_per_trade': profit_per_trade,
        'mean_absolute_error': mean_absolute_error,
        'profitable_trades': profitable_trades,
        'total_trades': len(actual_prices),
        'buy_profits': buy_profits,
        'sell_profits': sell_profits
    }
```

Output Files Generated:

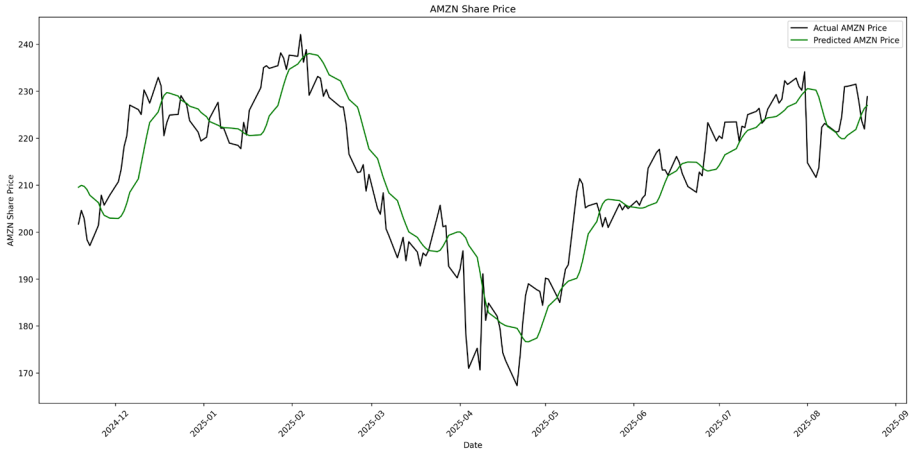
- CSV results: CBA.AX_next_day_20250817_205649.csv

```
Future price after 1 day is 224.91$
mean_squared_error loss: 0.0063705481588840485
Mean Absolute Error: 5.0741383622332314
Accuracy score: 0.5365853658536586
Total buy profit: 24.769989013671875
Total sell profit: 7.9199981689453125
Total profit: 32.68998718261719
Profit per trade: 0.7973167605516387
```

- Performance plot: CBA.AX_plot.png



AMZ (for easier comparison with P1)



2.2 P1 Testing Results

Fixes Applied:

Fixed data loading function using yfinance

```
def get_stock_data(ticker):
    """
    Get stock data using yfinance (modern version of yahoo_fin).
    """
    # Default to 2 years of data
```

```
    end_date = datetime.now()
    start_date = end_date - timedelta(days=730)
    data = yf.download(ticker, start=start_date, end=end_date, progress=False)

    if data is not None and not data.empty:
        # Handle multi-level columns from yfinance
        if isinstance(data.columns, pd.MultiIndex):
            # Flatten the column names
            # yfinance sometimes returns data with multi-level col names like:
            # ('Close', 'AAPL')
            # ('Volume', 'AAPL')
            # ('High', 'AAPL')
            # => We want to flatten them to: ['close', 'volume', 'open', 'high', 'low']
            data.columns = [col[0].lower() if col[1] == ticker else f"{col[0].lower()}_{col[1].lower()}"
for col in data.columns]
            # Rename 'close' to 'adjclose' to match expected format
            if 'close' in data.columns:
                data = data.rename(columns={'close': 'adjclose'})
        print(f"Successfully loaded {ticker} data using yfinance")
        return data
    else:
        raise ValueError(f"Failed to load data for {ticker}")
```

Fixed loss function compatibility

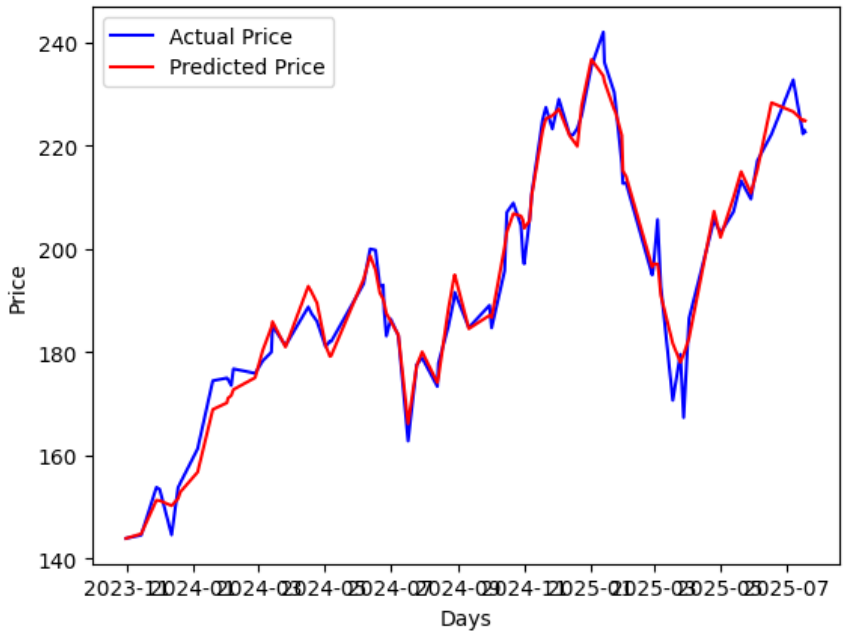
```
LOSS = "huber" # change from “huber_loss”
```

Performance Results:

```
Future price after 15 days is 223.07$
huber: 0.00044861173955723643
Mean Absolute Error: 2.7664683948863638
Accuracy score: 0.9545454545454546
Total buy profit: 559.6899871826172
Total sell profit: 355.78997802734375
Total profit: 915.4799652099609
Profit per trade: 10.403181422840465
```

Output Files Generated:

- Model weights: Saved to p1/results/
- **CSV results:** p1_output.csv
- Performance visualization:



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3. Understanding of v0.1 Code Base

3.1 Architecture Overview

The v0.1 implementation represents a basic LSTM-based stock prediction system with the following characteristics:

Core Components:

1. **Data Loading:** Yahoo Finance via yfinance library
2. **Preprocessing:** MinMaxScaler normalization (0-1 range)
3. **Sequence Creation:** 60-day sliding windows
4. **Model Architecture:** Stacked LSTM with dropout layers
5. **Prediction:** Single next-day price forecast

Model Structure:

```
model = Sequential([
  LSTM(units=50, return_sequences=True, input_shape=(60, 1)),
  Dropout(0.2),
  LSTM(units=50, return_sequences=True),
  Dropout(0.2),
  LSTM(units=50),
```

```
    Dropout(0.2),
    Dense(units=1)
])
```

3.2 Algorithm Flow

Step-by-Step Process:

1. **Training Data Collection:** Downloads historical stock data (2020-2023)
2. **Training Data Normalization:** Scales ONLY training data closing prices to [0,1] range using MinMaxScaler fitted on training data
3. **Sequence Generation:** Creates 60-day input sequences from training data to predict next day
4. **Model Training**: Trains LSTM network for 25 epochs with batch size 32
5. **Test Data Processing:** Downloads separate test period data (2023-2024)
6. **Test Data Scaling:** Applies same scaler (fitted on training data) to test data
7. **Prediction:** Generates predictions and inverse transforms using same training scaler.

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4. Performance Comparison

4.1 v0.1 and P1 comparison

Aspect	**v0.1 Approach**	**P1 Approach**	**Analysis**
Stock Symbol	AMZN (Amazon Inc.)		Identical for fair comparison
Data Period	Last 730 days (dynamic)		Identical timeframe
Train/Test Split	Sequentially, load train and test data separately	Randomly (configurable: 80%/20%)	**v0.1** is more manual
Scaling Strategy	Training data only (prevents test leakage)	All data together (test value range leakage)	**v0.1** theoretically correct
Prediction Horizon	1 day ahead	15 days ahead	**P1** has longer predictions horizon
Features Used	1 (Close price only)	5 (Open, High, Low, Close, Volume)	**P1** has 5x more information
Model Architecture	**Simple** LSTM (3 layers, 50 units)	More **complex** LSTM (2 layers, 256 units)	**P1** more sophisticated
Training Epochs	25	500	**P1** trains 20x longer
Training Loss	0.00637 (mse)	0.00044 (huber)	**P1** achieves 14x lower training loss
Mean Absolute Error	5.074 %	2.766 %	**P1** has lower MAE
Accuracy Score	53.7 %	95.5 %	**P1** has 78% improvement
Total Buy Profit	$24.77	$559.69	**P1** is 23x better in buy profit
Total Sell Profit	$7.92	$355.79	**P1** is 44x better in sell profit
Total Profit	$32.69	$915.48	**P1** is 28x more profitable

Profit per Trade	$0.797	$10.403	**P1** has 13x increase in profit for each trade

4.2 Comparison breakdown

Visual Analysis:

- **v0.1** has**:**
  - Shows prediction delays when following market trends.
  - Smoother predictions, which miss out critical peaks and valleys.
  - Underestimates peaks during upward trends
  - Overestimates peaks during downward trends.
  - Less responsive behavior to market volatility.

- **p1:**
  - More responsive prediction to the market trend
  - Follow the trend better but still miss some peaks.

Note: Visual comparison has limitations due to different splitting techniques: P1 has random train/test split while is v0.1 has sequential.

P1 has **better prediction accuracy** (95.5% vs 53.7%) and **trading profitability** ($915.48 vs $32.69 total profit). However, **v0.1** offers **advantages in computational efficiency** with 20x faster training (25 vs 500 epochs) and theoretically sound data handling that prevents information leakage. The trade-off between accuracy and speed makes P1 optimal for production trading systems, while v0.1 suits rapid prototyping and resource-constrained environments.

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5. GitHub Repository Setup

Highlight structure for the Github Repository:

```
stock-prediction /
├── v0.1/                   # Original YouTube tutorial code (fixed)
│   ├── v0.1.py             # Fixed stock prediction implementation
│   └── results/            # Output files and results
├── p1/                     # GitHub project implementation
│   ├── p1.py               # Core functions and data processing
│   ├── train.py            # Model training script
│   ├── test.py             # Model evaluation script
│   ├── parameters.py       # Configuration parameters
│   ├── p1.ipynb            # Jupyter notebook version
│   └── results/            # Model outputs and results
├── dev/                    # For task 2
├── utils/                  # Shared utilities (evaluating_data, plots)
├── requirements.txt        # Package dependencies
└── README.md               # This file
```

Appendix

Files run successfully

V0.1:

```
(venv) (base) PS C:\Users\User\OneDrive - Swinburne University\IS\assignment1\code> python v0.1/v0.1.py
2025-08-17 20:56:02.797967: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results
 due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=
0`.
2025-08-17 20:56:04.851410: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results
 due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=
0`.
C:\Users\User\OneDrive - Swinburne University\IS\assignment1\code\v0.1\v0.1.py:51: FutureWarning: YF.download() has changed argument auto_adjust
default to True
  data = yf.download(COMPANY,TRAIN_START,TRAIN_END)
[*********************100%**********************]  1 of 1 completed
2025-08-17 20:56:07.318 | INFO     | __main__:<module>:52 -
data
2025-08-17 20:56:07.321 | INFO     | __main__:<module>:68 -
2D scaled data
2025-08-17 20:56:07.321 | INFO     | __main__:<module>:92 -
1D scaled data
2025-08-17 20:56:07.323 | INFO     | __main__:<module>:102 -
x_train
2025-08-17 20:56:07.323 | INFO     | __main__:<module>:103 -
y_train
2025-08-17 20:56:07.323 | INFO     | __main__:<module>:108 -
x_train
2025-08-17 20:56:07.326478: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instruc
tions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flag
s.
C:\Users\User\OneDrive - Swinburne University\IS\assignment1\code\venv\Lib\site-packages\keras\src\layers\rnn\rnn.py:199: UserWarning: Do not pas
s an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the
 model instead.
  super().__init__(**kwargs)
Epoch 1/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 3s 28ms/step - loss: 0.0659
Epoch 2/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - loss: 0.0102
Epoch 3/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step - loss: 0.0078
Epoch 4/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - loss: 0.0078
Epoch 5/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step - loss: 0.0067
```

```
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 29ms/step - loss: 0.0055
Epoch 22/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 27ms/step - loss: 0.0049
Epoch 23/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - loss: 0.0051
Epoch 24/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - loss: 0.0052
Epoch 25/25
27/27 ━━━━━━━━━━━━━━━━━━━━ 1s 27ms/step - loss: 0.0049
C:\Users\User\OneDrive - Swinburne University\IS\assignment1\code\v0.1\v0.1.py:198: FutureWarning: YF.download() has changed argument auto_adjust
 default to True
  test_data = yf.download(COMPANY,TEST_START,TEST_END)
[*********************100%**********************]  1 of 1 completed
2025-08-17 20:56:28.651 | INFO     | __main__:<module>:200 -
test_data
2025-08-17 20:56:28.651 | INFO     | __main__:<module>:208 -
actual_prices
2025-08-17 20:56:28.651 | INFO     | __main__:<module>:211 -
total_dataset
2025-08-17 20:56:28.651 | INFO     | __main__:<module>:217 -
model_inputs
2025-08-17 20:56:28.651 | INFO     | __main__:<module>:220 -
model_inputs reshaped
2025-08-17 20:56:28.651 | INFO     | __main__:<module>:246 -
x_test
8/8 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
2025-08-17 20:56:49.715 | INFO     | __main__:<module>:283 - Prediction: [[115.16059]]

Next Day Prediction: $115.16
Last Known Price: $122.22
Predicted Change: $-7.06
Error: 5.8%
CSV saved: v0.1/results/CBA.AX_next_day_20250817_205649.csv
```

P1:

```
Epoch 499: val_loss did not improve from 0.00044
6/6 ━━━━━━━━━━━━━━━━━━━━ 1s 178ms/step - loss: 9.5774e-04 - mean_absolute_error: 0.0334 - val_loss: 8.1284e-04 - val_mean_a
bsolute_error: 0.0331
Epoch 500/500
6/6 ━━━━━━━━━━━━━━━━━━━━ 0s 139ms/step - loss: 0.0010 - mean_absolute_error: 0.0342
Epoch 500: val_loss did not improve from 0.00044
6/6 ━━━━━━━━━━━━━━━━━━━━ 1s 172ms/step - loss: 0.0010 - mean_absolute_error: 0.0342 - val_loss: 5.7745e-04 - val_mean_absol
ute_error: 0.0256
(venv) PS C:\Users\User\OneDrive - Swinburne University\IS\assignment1\code> python p1/test.py
2025-08-17 21:28:12.630172: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly diffe
rent numerical results due to floating-point ro
environment variable `TF_ENABLE_ONEDNN_OPTS=0`
2025-08-17 21:28:14.730394: I tensorflow/core/u
rent numerical results due to floating-point ro
environment variable `TF_ENABLE_ONEDNN_OPTS=0`
C:\Users\User\OneDrive - Swinburne University\I
ument auto_adjust default to True
  data = yf.download(ticker, start=start_date,
Successfully loaded AMZN data using yfinance
2025-08-17 21:28:17.426561: I tensorflow/core/p
 available CPU instructions in performance-crit
To enable the following instructions: SSE3 SSE4
ropriate compiler flags.
C:\Users\User\OneDrive - Swinburne University\I
serWarning: Do not pass an `input_shape`/`input
put(shape)` object as the first layer in the mo
  super().__init__(**kwargs)
C:\Users\User\OneDrive - Swinburne University\I
: UserWarning: Skipping variable loading for op
8 variables.
  saveable.load_own_variables(weights_store.get
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 106ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step
Future price after 15 days is 217.92$
huber loss: 0.00043791395728476346
Mean Absolute Error: 122.28282695354831
Accuracy score: 0.9310344827586207
Total buy profit: 522.1900329589844
Total sell profit: 348.44993591308594
Total profit: 870.6399688720703
Profit per trade: 10.007355964046786
```